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Research Article

Facile Synthesis of Lignin-Castor Oil-Based Oleogels as Green Lubricating Greases with Excellent Lubricating and Antioxidation Properties

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ABSTRACT: In this study, a facile synthesis method was employed to create lignin-castor oil-based oleogels by modifying organic lignin with two silane coupling agents. The resulting oleogels demonstrated outstanding lubricating and antioxidation properties, establishing them as promising green lubricating greases. Compared with the pure castor oil, the oxidation induction time (OIT) value of the synthesized oleogels was significantly increased from 20 s (pure castor oil) to 1959 s (oleogel with 20 wt % lignin), indicating an effective improvement of the oxidation resistance. The steel contacts lubricated by the synthesized oleogel also had lower wear than those lubricated by pure castor oil, signifying the better lubricating properties of oleogels. The oleogel with 20 wt % lignin showed the lowest wear, which was around 64% lower than that of pure castor oil. The exceptional performance and environmentally sustainable composition of these oleogels, with the biomass content exceeding 80%, allow them to be used as green lubricating greases for industrial applications. Overall, this study provides valuable insights into the development of high-performance, eco-friendly lubricants.



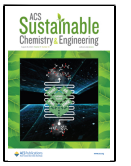
KEYWORDS: *lignin, castor oil, antioxidant, antiwear, oleogel*

INTRODUCTION

Lubricants play an important role in sustainable movements.[1] Roughly 80% of machine components fail due to wear, with friction and wear-related economic losses accounting for 2−7% of each nation's annual gross domestic product. Friction consumes about one-third of the world's main energy.[2] The presence of lubricants is vital in all mechanical systems as lubricants reduce the wear and friction of moving parts by creating an appropriate oil film between the interface of metals. Grease and lubricating oil are two of the common lubricants used in industry. Grease is generally used on metal surfaces to fill voids and prevent rust. Compared to lubricating oil, grease has a higher load-carrying capacity and better damping and shock absorption ability. Grease has excellent properties when operating at high temperatures and for extended periods.[3] Only a small amount of grease is required to lubricate a surface, significantly reducing the requirement for lubricating oil.[3,4] This is undoubtedly of great significance for today's energy shortage. However, due to the large amount of mineral oil utilized, nearly 50% of it enters the environment and comes into direct contact with water and soil, causing irreparable damage to the environment.[5] The use of unsustainable fossil raw materials also limits the development of grease.[6]

During the past few decades, green lubricants, which are nontoxic and have little impact on the environment, have gained increasing attention in addressing the impact of petroleum-based lubricants on the environment.[7] Lubricants based on vegetable oils or animal fats are considered as green lubricants and have gained popularity and acceptance due to their sustainable and environmentally friendly properties.[8] Green grease can also be created by mixing vegetable oil or animal fat with a green thickener.[9] Several such green thickeners have been reported, such as methylated cellulose pulp, carboxymethyl guar gum, sorbitan or glyceryl monostearate, and modified lignin.[10−13] Modified lignin has become one of the most promising green thickeners due to its broad range of sources, excellent antioxidant properties, and biodegradable properties.[14,15] In the work done by Borrero-López et al.,[16] different diisocyanates (hexamethylene

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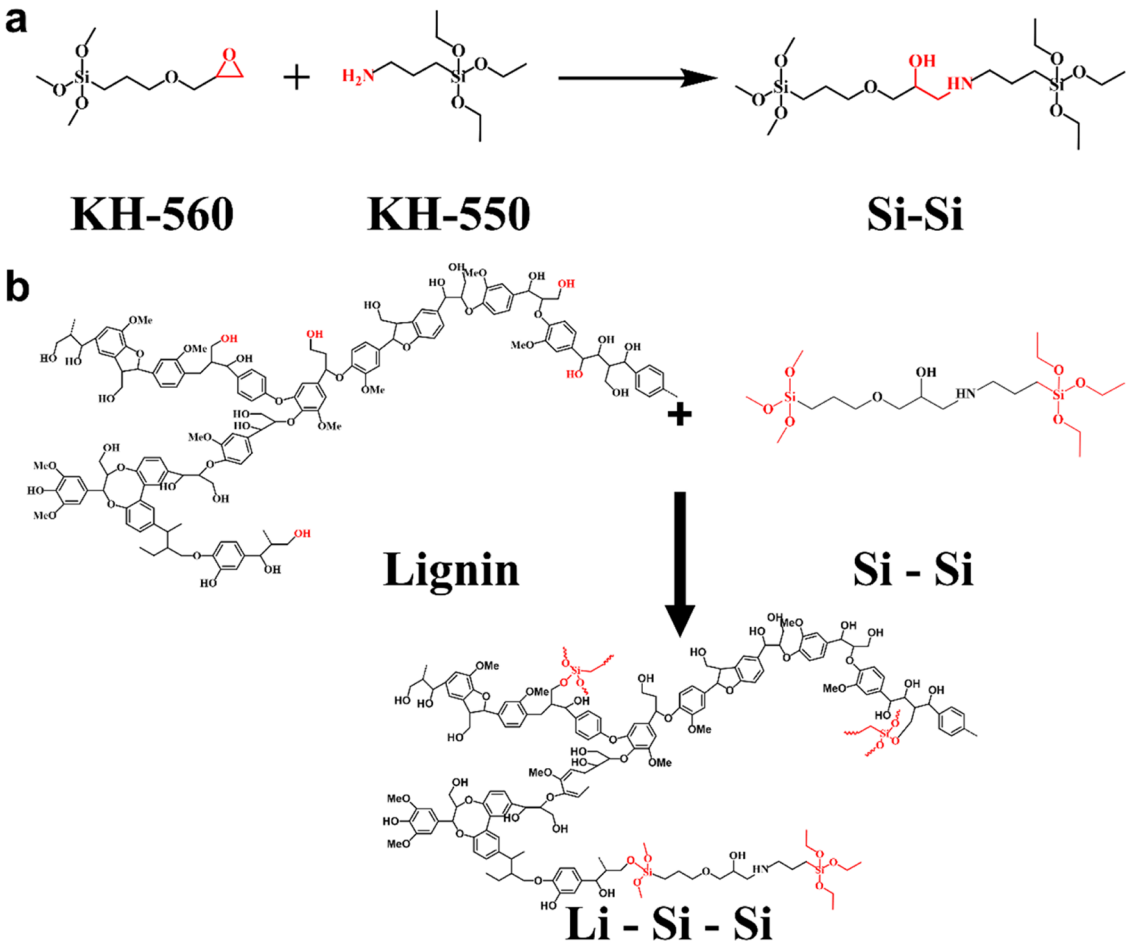


Figure 1. (a) Reaction scheme of KH-550 and KH-560. (b) Reaction scheme of lignin and Si−Si.

diisocyanate (HDI), isophorone diisocyanate (IDI), toluene diisocyanate (TDI), and 4,4′-methylene bis (phenyl isocyanate) (MDI))-modified alkali lignins have been tested as gelling agents in castor oil medium. In another work, researchers functionalized the side chains of different lignins with HDI in a toluene solution and used them as a thickener in castor oil to produce oleogels.[17] Although most of the raw materials of this grease come from natural products, the toxic isocyanate makes the production process not so environmentally friendly.[18] On the other hand, the authors did not study the wear resistance and oxidation properties of the developed lignin-based oleogels.

As a natural phenolic compound found in plant biomass, lignin is the second most abundant biopolymer material on earth. Lignin incorporation into polymers has been widely explored by researchers due to its antimicrobial properties, UV shielding, antioxidant activity, thermal stabilization, lubricity, and other properties.[19−22] As one of the three main components of lignocellulose, lignin is the only aromatic biopolymer. Depending on the applied isolation method, lignin can be categorized into different types, such as acid lignin, alkali lignin, and organosolv lignin.[23−25] The way lignin is extracted affects its final chemical properties as the bond cleavage and formation occur at varying degrees during the process. The organic solvent route for extracting lignin can greatly reduce pollution by utilizing the good solubility and volatility of organic solvents. The organic solvent method is becoming a very promising extraction and separation method for establishing biorefinery processes.[26] Paulsen Thoresen et al.[27] studied the effect of process parameters of organosolv fractionation of silver birch and established a correlation between the process parameters of organosolv and the structural characteristics of the isolated lignins for application-specific lining production.

Lignin modification can considerably improve the utilization rate of lignin. Zhang et al.[28] introduced butyric acid-modified lignin into the synthesis process of polyurethane, improving the mechanical and dielectric properties of polyurethane. Zhou et al.[29] grafted polybutylene dipropionate on lignin and successfully synthesized a green polybutylene dipropionate-lignin gel compatibilizer with a new network structure.

In this study, we propose a novel and facile approach to modify organosolv lignin, by reacting with two silane coupling agents. The modified lignin was further reacted with castor oil to form a new type of oleogel that showed outstanding lubricating and antioxidation properties. This facile synthesis method and the use of renewable resources make the oleogel a green lubricant for future industrial applications. Overall, this study provides valuable insights into the development of green lubricants with improved performance and sustainability for various applications.

MATERIALS AND METHODS

Material. Beechwood (LIGNOCEL HBS 150/500; Rettenmaier Sweden KB JRS), sulfuric acid (H_2SO_4; 95.0−98.0%; Sigma-Aldrich), ethanol (max. 0.01% H_2O, Sigma-Aldrich), (3-aminopropyl)

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triethoxysilane (KH-550; ≥98%; Sigma-Aldrich), (3-glycidyloxypropyl) trimethoxysilane (KH-560; ≥97%; Sigma-Aldrich), acetone (≥99.8%, Sigma-Aldrich), castor oil (AR; Shanghai Macklin), LGWM 2/0.4 grease (SKF), Teboil MultiPurpose Grease (Oy Teboil Ab). All chemicals and materials were used as received without further treatment.

Extraction of Lignin. Organosolv lignin was isolated from beechwood. In short, the lignocellulosic material was processed in an air-heated multidigester system (Haato, Vantaa, Finland) in 60/40 vol % ethanol/water with 1 wt % H_2SO_4 (on dry biomass basis), at 180 °C and for 60 min, and at a solid/liquid ratio of 1:10 g/mL. After the organosolv treatment, the generated slurry was vacuum-filtered and the liquid fraction, obtaining dissolved lignin and released sugar-based components, was processed by evaporation (rotary evaporation; Heidolph, Schwabach, Germany) to reduce the solubility of lignin. After lignin precipitation, this fraction was recovered by centrifugation at 10 000*g* for 15 min at 4 °C (5804R; Eppendorf, Hamburg, Germany) and freeze-dried (Lyoquest; Telstar, Terrassa, Spain) before being stored in plastic flasks in room temperature.

Oleogel Synthesis. The process of preparing modified lignin-based oleogel involved a three-step approach. First, 4.7 g of KH-550 was reacted with 5 g of KH-560 at room temperature for 12 h at a molar ratio of 1:1 to obtain the product Si−Si, as depicted in Figure 1a. Next, Si−Si was reacted in different proportions with organosolv lignin in acetone at room temperature for 12 h to produce the thickener Li−Si−Si. The reaction schematic diagram for producing Li−Si−Si is shown in Figure 1b. Finally, Li−Si−Si was then added to a mixture of castor oil and acetone in the ratios outlined in Table 1,

Table 1. Feed Ratio of Lignin-Based Oleogels

	castor oil	10% lignin	15% lignin	20% lignin	25% lignin	30% lignin
550−560 (g)	0	0.96	1.46	1.87	2.26	2.74
organosolv lignin (g)	0	0.32	0.45	0.63	0.76	0.92
castor oil (g)	12	11.5	11	10	9	8.5

with the solid content (Li−Si−Si and castor oil) controlled at 30 wt %. The mixture was blended at room temperature for 12 h to allow a good reaction between Li−Si−Si and castor oil. The final product (lignin-based oleogel) was received by placing the mixture in a blast oven at 100 °C for 6 h to remove the solvent. The reaction schematic diagram for producing lignin-based oleogels is shown in Figure 2.

Fourier Transform Infrared Spectroscopy. Fourier transform infrared—attenuated total reflectance (FTIR-ATR) spectra were recorded using a Nicolet Summit FTIR spectrometer. The number of scans is 16, and the resolution is 4 cm. The wavenumber range was determined to be 4000−400 cm^{-1}.

Thermogravimetric Analysis. The castor oil and lignin-based oleogel samples were dried at 105 °C prior to thermogravimetric analysis (TGA). TGA was carried out using a TGA 8000-PerkinElmer, with an integrated thermal analyzer, nitrogen flow rate (100 mL/min), heating rate (10 K/min), and a test temperature range from room temperature to 600 °C.[14]

Rheology Study. The GD-4929A drop point tester from Chongqing Gold Mechanical & Electrical Equipment Co., Ltd. was employed to conduct the drop point test according to the ASTM D566 and GB/T 4929 standards. The test was performed by gradually heating the drop point tester at a constant ramp rate of 5 °C per minute. Each sample was tested three times with two parallel comparison samples.

To determine the rheological properties of the castor oil and lignin-based oleogels, experiments were carried out by using an Anton Paar rheometer, SmartPave 92, with a plate-plate configuration (20 mm diameter, 1 mm gap). Initially, dynamic temperature sweep mode was used with a 5% strain within the temperature range of 20−80 °C (the temperature range is determined based on drop point test results) to obtain the storage modulus (G') and loss modulus (G''). The dynamic temperature sweep mode was used to examine the viscosity and shear stress with a temperature range of 20−80 °C. The shear stress and viscosity of various samples were tested by subjecting them to a constant strain of 0.1% and a shear rate range of 0.01−100 ($1/s$) at a temperature of 25 °C.

Tribology Study. To ensure the standardization of the lubricating performance tests, the parameters used in this study conformed to the ASTM D5707-11 standard. The tests were conducted on an SRV tribometer (Optimol Instruments, Munich, Germany) with an upper steel ball (52 100 bearing steel, diameter 10 mm, surface roughness (Ra) 20 nm) undergoing reciprocating motion against a stationary steel disk (100CR6 ESU hardened, Ø24 mm × 7.9 mm, surface roughness (Ra) 120 nm). The tests were run for a duration of 1 h under a contact pressure of 2.72 GPa at a temperature of 80 °C, with a sliding frequency of 50 Hz and 1 mm amplitude. The experiment was repeated three times to ensure the reproducibility of the results. The lower disk wear volume loss and the upper ball worn diameter were determined by using a 3D optical profile meter (Zygo 7300) after the tribology tests.

Antioxidation Study. Castor oil and lignin-based oleogels were stirred in the air at 180 °C for 3 h. The purpose of stirring is to fully contact the sample with oxygen. The samples before and after the oxygen resistance test were tested by FTIR spectroscopy, respectively, at 0, 20, and 60 min and 2 and 3 h during the test process.

Oxidation induction time (OIT) was determined by DSC (TA Instruments DSC Q2000). Samples (5−10 mg) were exposed to a nitrogen atmosphere with a nitrogen flow rate of 50 mL/min and a heating rate of 20 °C/min. After reaching the temperature of 200 °C, the samples were kept in a nitrogen atmosphere for 10 min. The nitrogen flow was then switched to an oxygen flow of 50 mL/min and maintain the temperature until the sample was completely oxidized. OIT was reported as the onset time of sample oxidation.

RESULTS AND DISCUSSION

FTIR Analysis. The FTIR spectra of KH-550, KH-560, and Si−Si are shown in Figure 3a. The peak at 3300−3500 cm^{-1} is the asymmetric stretching vibration of the primary amino group of KH-550,[30] while the peak at around 1600 cm^{-1} is the asymmetric bending vibration peak of the primary amino group of KH-550. The infrared characteristic absorption peak of the epoxy group of KH-560 appears at 910 cm^{-1} nearby.[31] The epoxy ring in KH-550 and KH-560 undergoes a nucleophilic addition reaction with the amino group to open the epoxy ring and generate a product with two hydroxyl functional groups.[32] As the reaction time increased, the characteristic peaks of the primary amino group and the epoxy group gradually disappear, indicating that KH-550 and KH-560 are connected because of the reaction of the primary amino group and epoxy group (Figure S3).

It can be seen from Figure 3b that the peak at 918 cm^{-1} is the characteristic peak of the epoxy five-membered ring. This peak appears at the same position for lignin−Si−Si. At the same time, the silane coupling agent was first hydrolyzed in acetone to generate Si−OH, and then dehydrated with the hydroxyl groups on the lignin surface to form Si−O−C covalent bonds. A new peak was generated at 1080 cm^{-1}, which is the characteristic peak of Si−O−C.[33] This shows that lignin reacts with Si−Si. The characteristic peak of Si−O−C appeared on the infrared spectrum for lignin-castor oil. The FTIR studies indicate that the synthesis of oleogels is consistent with the designed route.

Thermogravimetric Analysis. The thermal decomposition of the samples was investigated by TGA, which indirectly reflected the effect of modified lignin on the thermal stability of the oleogel. Figure 4b shows the thermal decomposition diagram of castor oil and lignin-based oleogel through the thermal decomposition curve (DTG). It can be seen in Figure

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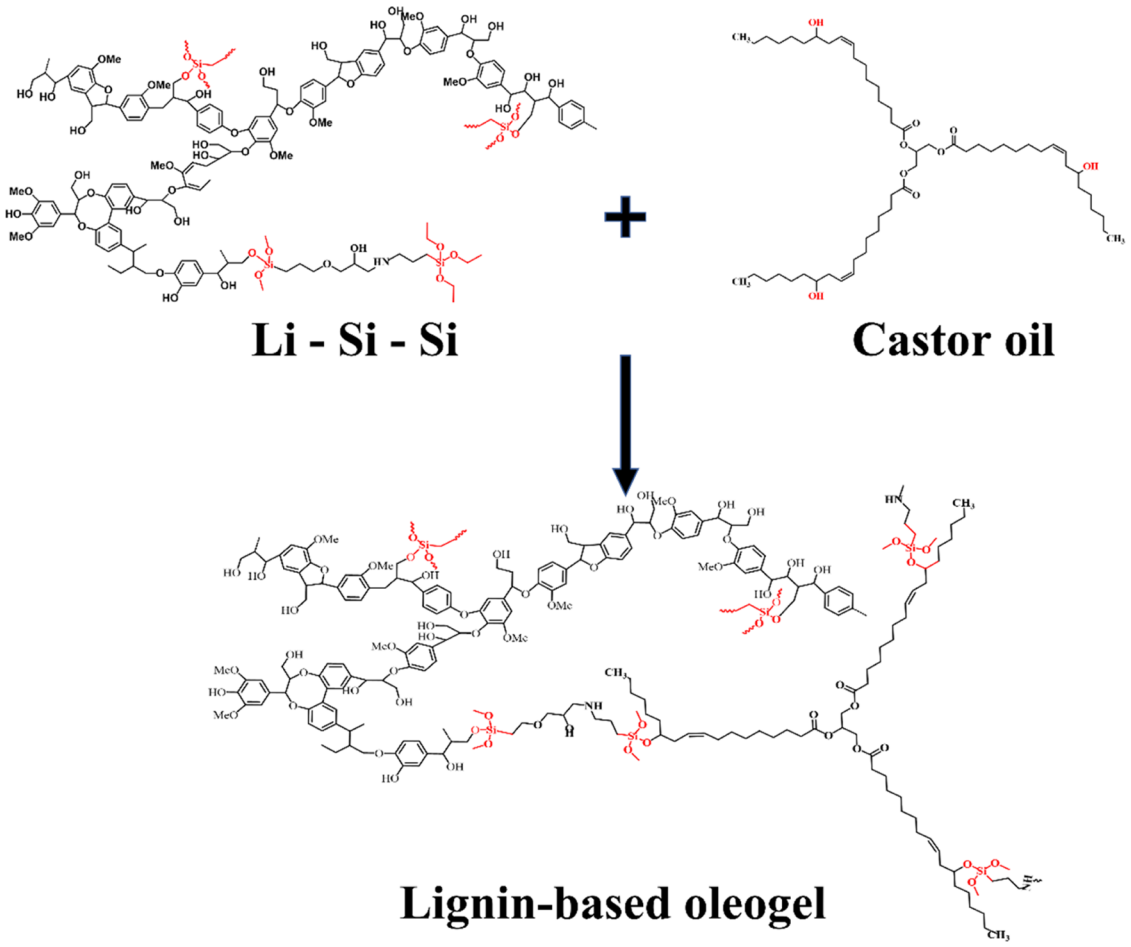


Figure 2. Reaction schematic diagram of Li−Si−Si and castor oil.

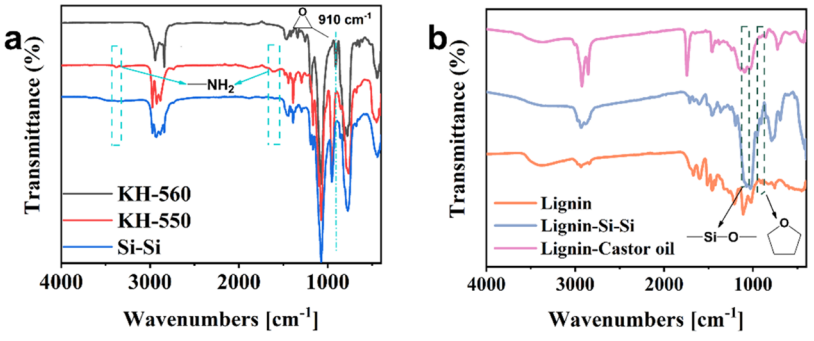


Figure 3. (a) FTIR spectra of KH-550, KH-560, and Si−Si. (b) FTIR analysis of lignin, modified lignin, and lignin-based oleogel.

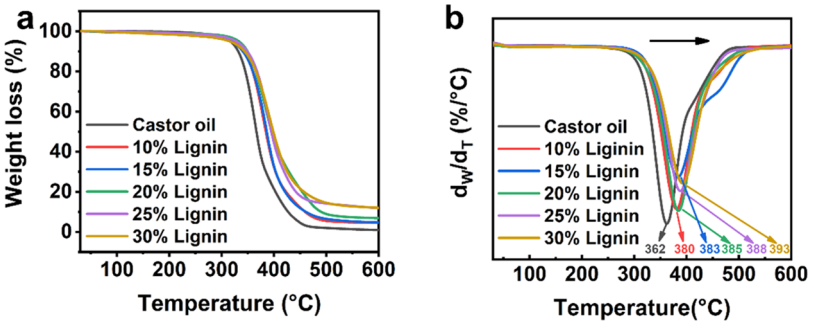


Figure 4. (a) TG and (b) DTG of castor oil and lignin-based oleogel.

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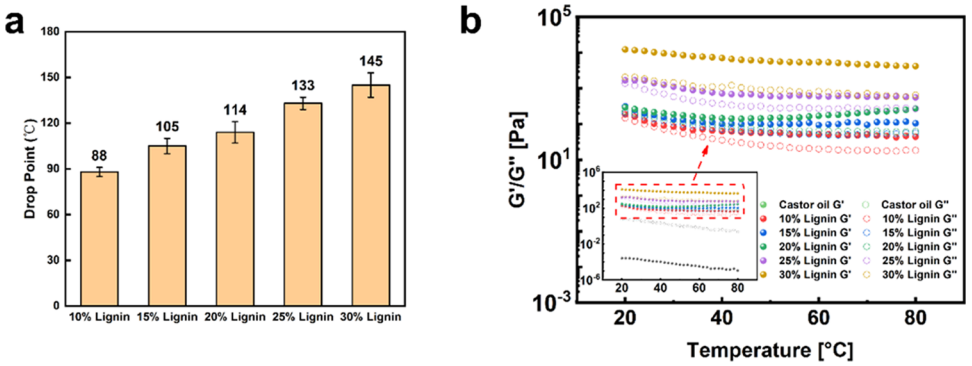


Figure 5. (a) Drop point of lignin-based oleogels. (b) Temperature-dependent plots of storage and loss moduli of castor oil and lignin-based oleogels.

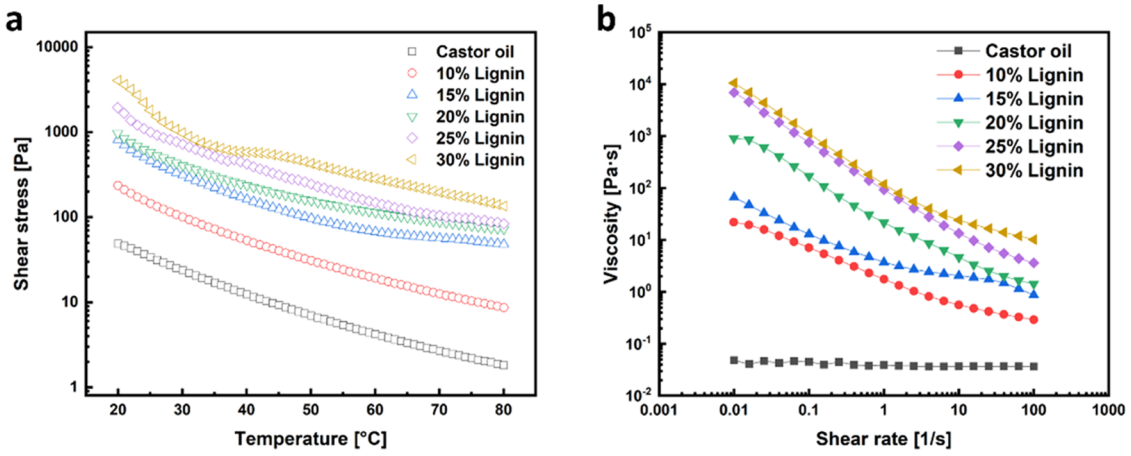


Figure 6. (**a**) Plot of shear stress as a function of temperature. (**b**) Viscosity as a function of shear rate at different thickener concentrations at 25 °C.

4a that with the increase of the content of modified lignin, the initial decomposition temperature of the sample was increased from 275 to 291 °C, and the weight loss of the sample decreased from 99.02% of castor oil to 87.89% of 30% lignin-based oleogel. At the same time, it can also be found from DTG (Figure 4b) that the decomposition temperature of the sample also increased from 362 to 395 °C. This indicates that the addition of modified lignin in the sample improves the thermal decomposition resistance of the oleogel. This can be attributed to two reasons: (1) New hydrogen bonds were formed between the modified lignin and castor oil. Yuan et al. found that an increase in the number of hydrogen bonds in the colloid could enhance the thermal stability.[34] So, the increased hydrogen bonds between lignin and castor oil would have the same impact and increase the initial decomposition temperature. (2) As a chemical thickener, the modified lignin forms a strong chemical bond −Si−O− inside the oleogel, thereby enhancing the thermal stability of the oleogel. The thermal decomposition temperature of lignin-based oleogel is influenced by the distribution of hydrogen bonds at the molecular level as well as the molecular weight distribution. As we reported in our previous work on PEG/lignin lubricant, the thermal decomposition temperature is synergistically affected by hydrogen bonding and molecular weight distribution.[35]

Rheological Behavior. The dropping point refers to the lowest temperature at which the oleogel starts to drip during heating, which is one of the important indicators of an oleogel.[36] It is often desirable to obtain a high drop point because a high drop point means a wide range of applications. It can be found from Figure 5a that with the increase of lignin content, the drop point of the oleogel increases from 88 to 145 °C. This is due to the formation of more intermolecular hydrogen bonds inside the oleogel with the increase of modified lignin. At the same time, the dropping point of the oleogel also provides a reference for the subsequent rheological studies.

Storage modulus (G'), also known as elastic modulus, refers to the amount of energy stored due to elastic (reversible) deformation when a material is deformed, reflecting the elasticity of the material.[4] The loss modulus (G'') is a measure of its viscoelasticity. G'' represents the energy lost in the form of heat when a grease is sheared or stressed.[37] It can be found from Figure 5b that the G'' of castor oil is much larger than G' in the temperature range of 20−80 °C, indicating that the material mainly undergoes viscous deformation, which means the material is liquid. For the oleogel, G' and G'' were close to each other. Although G' and G'' of the oleogel decreased with the increase of temperature, G' was always slightly higher than G'', showing that the oleogel was always semisolid. As the temperature increased, the intermolecular hydrogen bonds of the oleogel were gradually destroyed, resulting in gradually decreased G' and G''.[38] This decrease in modulus weakens the network structure and makes the oleogel behave more like a liquid. At low temperatures, intermolecular hydrogen bonding between the castor oil and the modified lignin molecules forms

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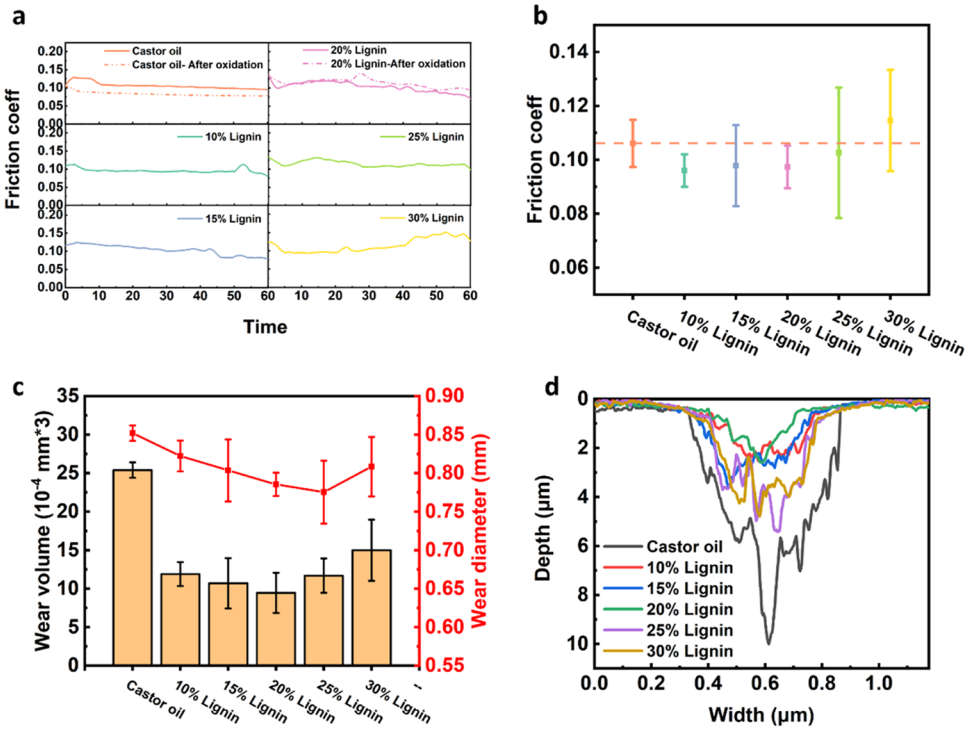


Figure 7. (a) Friction coefficient of castor oil, castor oil after oxidation, lignin-based oleogel, and 20% lignin after oxidation. Load: 2.76 GPa, testing duration: 1 h, temperature: 80 °C. (b) Values of the friction coefficient. (c) Disk wear volume lubricated by castor oil and lignin-based oleogel. Load: 2.76 GPa, testing duration: 1 h, temperature: 80 °C. (d) Cross-sectional picture of disk wear while lubricating by castor oil and lignin-based oleogel. Load: 2.76 GPa, testing duration: 1 h, temperature: 80 °C.

a strong network structure, which increases the G' of the oleogel and makes it semisolid.

As can be seen from Figure 6a, as the lignin content increases, the shear stress of the oleogels increases. This is due to the chemical bond between lignin and castor oil, which increases the cross-linking points of the lignin-based oleogels, thereby improving the cross-linking degree of the lignin-based oleogels. When the temperature increased, the shear stress of castor oil and lignin-based oleogels gradually decreased. It is not difficult to conclude that the addition of lignin introduces many hydrogen bonds, and when the temperature rises, these intermolecular hydrogen bonds are destroyed.[39]

It can be observed from the rheological analysis results (Figure 6b) that castor oil is a Newtonian fluid, as the relationship between shear stress and shear strain rate obeys Newton's law of internal friction or Hooke's law (Figure S5). In contrast, lignin-based oleogels are pseudoplastic fluids due to their internal irregular order at rest, resulting in higher viscosity (Figure 6b) and resistance to flow.[40] With the increase of shear rate, the influence of external stress on the intramolecular hydrogen bonds of lignin-based oleogel breaks, and the molecular chains are unwound, stretched, and oriented along the driving force direction. The aligned molecular coils can slide over each other more easily allowing the lignin-based oleogels to flow faster under a given shear stress.[4] This is also the reason why the viscosity of lignin-based oleogels decreases with increased shear rate (Figure 6b).

Tribological Performance Test. Ball wear diameter is an important antiwear property index for lubricant evaluation. Figure 7a shows the evolution of the coefficient of friction (μ) during 1 h of friction tested at a pressure of 2.76 GPa based on lignin oleogel with castor oil. A relatively high coefficient of friction (0.106) was observed when pure castor oil was used. The addition of lignin to castor oil had a significant positive effect on the reduction and stabilization of the friction coefficient during the friction test. When the content of modified lignin was 20%, the friction coefficient of lignin-based oleogel was the lowest (0.097) (Figure 7b). The addition of modified lignin to castor oil can create a protective layer on the metal surface, which acts as a barrier between the metal surfaces and reduces friction between them. This protective layer is formed through a process known as adsorption, where the modified lignin molecules attach to the metal surface and form a film. The film reduces the direct contact between the metal surfaces and acts as a cushion, reducing the friction between them. This claim has been proven in our previous studies.[41,42] The creation of this protective layer is particularly important in boundary lubrication, where the lubricating film is thin, and the surfaces are in direct contact. When the content of modified lignin is low, the lubricating effect is not significant enough to reduce the friction coefficient. On the other hand, when the content of modified lignin is too high, the excess lignin can form agglomerates that reduce the efficiency of the lubricant film, resulting in a higher friction coefficient (Figure S8).[43]

Figure 7c shows the wear volume loss (V) of the steel disk and the wear diameter (D) of the steel ball lubricated by castor oil and lignin-based oleogel. The wear volume loss of lignin-based oleogel was significantly lower than that of castor oil, indicating the positive contribution of lignin for enhancing the lubricating properties. When the modified lignin was added into castor oil as a thickener, the wear volume of the steel disk and the wear diameter of the steel ball obviously decreased (Figure 7d). The oleogel with 20% modified lignin as a thickener showed the lowest wear (around 64% lower than that of castor oil). When modified lignin is added to castor oil,

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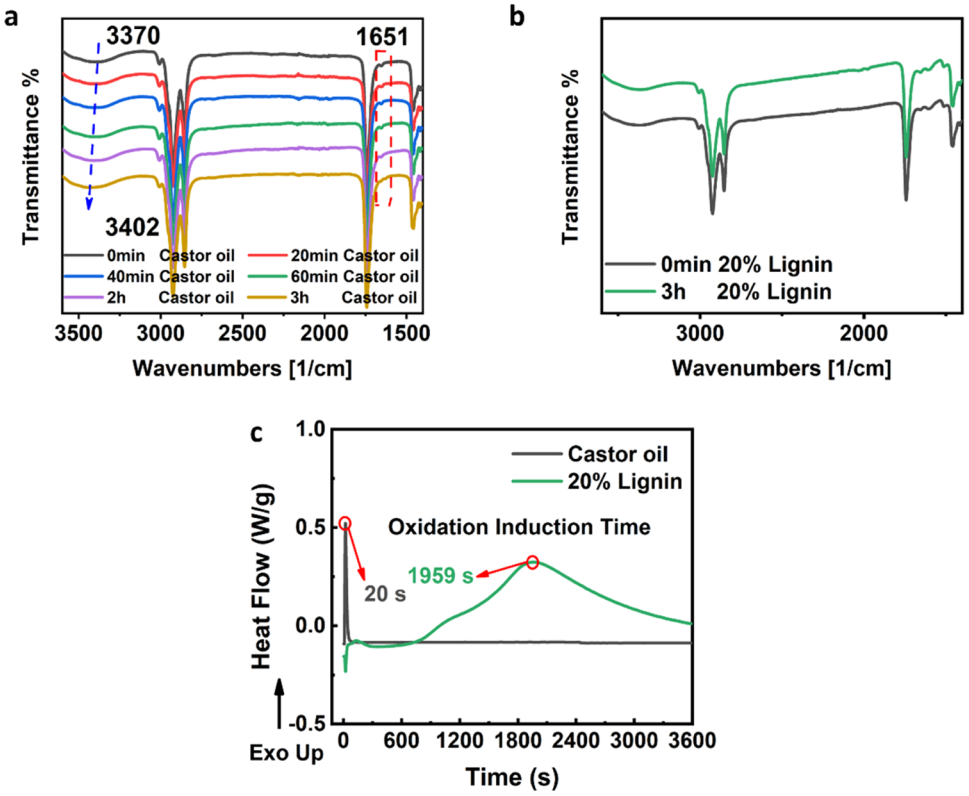


Figure 8. (a) Infrared change diagram of castor oil oxidized at 180 °C for 3 h. (b) Infrared contrast diagram of 20% lignin oleogel before and after oxidation at 180 °C for 3 h. (c) OIT results of castor oil and 20% lignin oleogel.

it can adsorb onto the metal surface, creating a protective layer.[42] This layer reduces direct metal-to-metal contact and minimizes wear and friction between the surfaces.[44] Additionally, the presence of modified lignin in the oleogels enhances the film strength of the lubricant, which further improves its antiwear properties.[35,45] But when the content of modified lignin continued to increase, the antiwear performance of lignin oleogel decreased. This is because the excess modified lignin will disrupt the network structure of the oleogel, leading to a decrease in its mechanical strength and stability, which will decrease its protection effect during rubbing.

It is worth noting that the comparison of lubricants revealed interesting findings. The 20% lignin-based oleogel exhibited a higher viscosity (6.4 Pa·s), a lower friction coefficient (0.103 ± 0.007), and a slightly larger wear mark diameter (0.785 mm) compared to the LGWM 2/0.4 grease and Teboil Multi-Purpose Grease (Table S1). It is important to highlight that the lignin-based oleogel does not contain any commercial antiwear additive or other lubricating additives that can effectively enhance the tribological properties. Despite this, the lignin-based oleogel demonstrates similar tribological performance to fully formulated commercial greases.

Antioxidation Test. The preservation of lignin-based oleogels from degradation and generation of harmful by-products, such as acids, sludge, and varnish, is vital and can be achieved through improving their antioxidation properties.[46] It is interesting to observe that the degradation does not obviously affect the lubricating performance (as shown in Figure 7a). However, it is well known that the anticorrosion and rheological properties will be greatly deteriorated, which will cause serious machine damages problems.[47]

To compare the antioxidant properties of castor oil and lignin-based oleogels, the oleogel contains 20% lignin with excellent friction and antiwear properties were selected for the antioxidant test together with castor oil. Two types of antioxidant tests were run; FTIR spectroscopy to analyze the occurrence of castor oil oxidation by changes in the chemical structure, and DSC to compare the oxidation resistance based on OIT.

The thermal oxidation of castor oil and lignin-based oleogels is influenced by the degree of unsaturation in the fatty acid chains. The presence of double bonds in the fatty acid chains creates reactive sites for oxidation in the form of free radicals.[48] The hydrogen atoms adjacent to these double bonds can be removed, generating lipid free radicals. These free radicals can further react with oxygen, forming lipid peroxyl radicals.[49] The lipid peroxyl radicals can then remove hydrogen atoms from neighboring fatty acid chains, causing a chain reaction that generates additional lipid free radicals and lipid hydroperoxides, which further catalyze the oxidation process.[50] Castor oil eventually degrades into low-molecular-weight volatile and nonvolatile chemicals.[51] It can be seen from Figure 8a that the double-bond peak of castor oil disappears and the −OH peak has a blue shift. The double bond of castor oil is destroyed in the high-temperature aerobic environment. However, the 20% lignin oleogel does not change before and after the oxidation experiment (Figure 8b).

The antioxidant effect of lignin-based oleogel was assessed in oxidative induction time (OIT) studies. Free radicals are produced on-site when materials are exposed to high-temperature oxygen in OIT experiments. By postponing the start of material oxidation, antioxidant benefits can be seen. Oil oxidation is an exothermic process that releases heat. It can be

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seen from Figure 8c that the oxidation induction time of castor oil in a pure oxygen environment at 180 °C is 20 s, while the oxidation induction time of 20% lignin oleogel is 1959 s. This is because the lignin extracted by organic solvent has a higher content of phenolic hydroxyl groups showing stronger antioxidant activity by breaking the ether bond (reducing the molecular weight).[S2]

CONCLUSIONS

In summary, this study presents a promising approach for the development of green lubricating greases using modified lignin and castor oil. The modified lignin and castor oil-based oleogels developed in this study had a high biomass content, as lignin is a byproduct of the pulp and paper industry, and castor oil is a renewable resource. The lignin modified by KH-550 and KH-560 is proved to be an effective thickener, which can effectively improve the antiwear and antioxidation properties. Specifically, the antiwear property of the oleogel is increased by 64% using 20% modified lignin as a thickener. Adding lignin-based thickener into castor oil can improve the hydrogen bond strength and thermal stability. The facile synthesis method and the use of natural and renewable resources allow the use of these oleogels as green lubricants for different applications. Overall, the findings of this study provide a significant contribution to the development of green lubricants that offer high performance and meet the growing demand for sustainable and renewable products.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssuschemeng.3c01801.

Disk wear pictures with castor oil and lignin-based oleogel; 3D microscopic images of ball wear trajectories; FTIR spectra of KH-550 and KH-560 reactions; DSC data for castor oil and lignin-based oleogel; viscosity and shear stress measurements at 25 °C; comparisons of coefficient of friction and OIT values for 20% lignin and two commercial greases; comprehensive analysis of viscosity and main lubricant properties for 20% lignin and two commercial greases; SEM analysis of modified lignin-based lubricants and hydroxyl content determined via 31P NMR (PDF)

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Notes

The authors declare no competing financial interest.

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